Physics of the Earth and Planetary Interiors, 34 (1984) 137–149
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

Palaeomagnetic evidence for the origin of Madeira

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(Received August 11, 1983; revision accepted December 19, 1983)

Abranches, M.C. and Storetvedt, K.M., 1984. Palaeomagnetic evidence for the origin of Madeira. Phys. Earth Planet. Inter., 34: 137–149.

The remanent magnetization of 'basement' volcanics from Madeira define three different axes of magnetization, each having a dual-polarity build-up. The suggested oldest of these components, with declination 302 and inclination +4, is assigned to the late Lower Cretaceous and is thought to reflect the age of the early volcanism of the island. Subsequent magnetization overprints seem to have occurred in the Late Cretaceous/Early Tertiary (minor) and in Neogene times, respectively. The latter magnetization, which is strongly developed, was most likely impressed during the extensive volcanism that swept the island in post-Late Miocene. The palaeomagnetic evidence for a Cretaceous origin of Madeira is supported by the finding of Lower–Middle Cretaceous tuff layers in DSDP site 136 which is located only 160 km north of the island. The inferred palaeomagnetic structure of the 'basement' rocks of Madeira is similar to that found in the old volcanic complexes of other east central Atlantic islands.

1. Introduction

The island of Madeira is located some 900 km southwest of Gibraltar and nearly 500 km north of the Canary Islands. In the general plate tectonic theory Madeira lies within the African plate, and according to the prevailing interpretation of marine magnetic anomalies, the crust in the Madeira region formed by sea-floor spreading at around 140 Ma ago (Pitman and Talwani, 1972). DSDP site 136 (Hayes et al., 1972) which is situated only 160 km north of the island (Fig. 1), ended in tholeiitic basalt (at 308 m subbottom) regarded as the topmost part of the true oceanic basement (layer 2), despite the fact that it is only ca. 18 m below recovered late Aptian nannoplankton marls. These sediments appear anomalously young in comparison with the suggested crustal age based on magnetic anomalies. Krejci-Graf (1964) has suggested that Madeira is geologically connected to the Upper Cretaceous volcanic province of western Portugal (cf. the linking ridge of Fig. 1, including Gorringe Bank). In the Canary and Cape Verde Islands, which have geological settings very similar

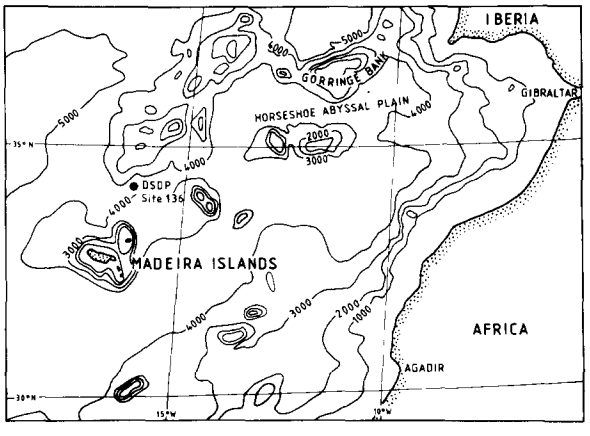


Fig. 1. Geographical setting of Madeira. The location of DSDP site 136 is marked by a solid circle.

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to that of Madeira, there is indeed growing evidence for Cretaceous volcanism (Storetvedt et al., 1978, 1979; Storetvedt, 1980; Stillman et al., 1982; Storetvedt and Løvlie, 1983). The present palaeomagnetic study has been undertaken, with the aim to provide further age information on the early volcanic activity of Madeira.

2. Geology and palaeomagnetic sampling

Madeira, which is a deeply dissected dome-shaped island, reaches maximum elevations of up to ca. 1850 m in its central region. Geological summaries (e.g., Gagel, 1912, 1914; de Morais, 1945, 1947; Krejci-Graf, 1961, 1964; Mitchell-Thomé, 1976) describe a practically undisturbed volcanic stratigraphy, consisting basically of lavas and pyroclastics. The oldest section, the 'basement' complex, is dominated by tuffs and volcanic breccias along with trachytic flows and minor alkali-rich intrusives. Higher up in the sequence lavas become more important, the youngest members (Pliocene–Pleistocene) being dominated by alkali-olivine basalts. Dense dyke swarms occur over most of the island.

In Ribeira S. Vincente (on the north coast) a poorly outcropping but richly fossiliferous reef deposit, now situated at an elevation of 360–400 m above sea level, lies on top of the oldest volcanic division. There is a general palaeontologic consensus in assigning a Late Miocene (Vindobonian) age to this limestone deposit. Other available age information comes from K/Ar data on a few lavas of the topmost volcanic sequences, giving Pliocene–Pleistocene radiometric ages (Watkins, 1973). The age of the 'basement' volcanics (i.e., pre-dating the Vindobonian sediments) has remained unknown.

A total of 45 oriented hand samples from 14 sites form the basis of this study. In the S. Vincente region the collected sites (nos. 5 and 6) are stratigraphically well below the local Vindobonian sediments. In fact, it is only the Lombado do Moleiro and S. Vincente sites (nos. 1–6) that clearly conform to the description of the 'basement' volcanics,

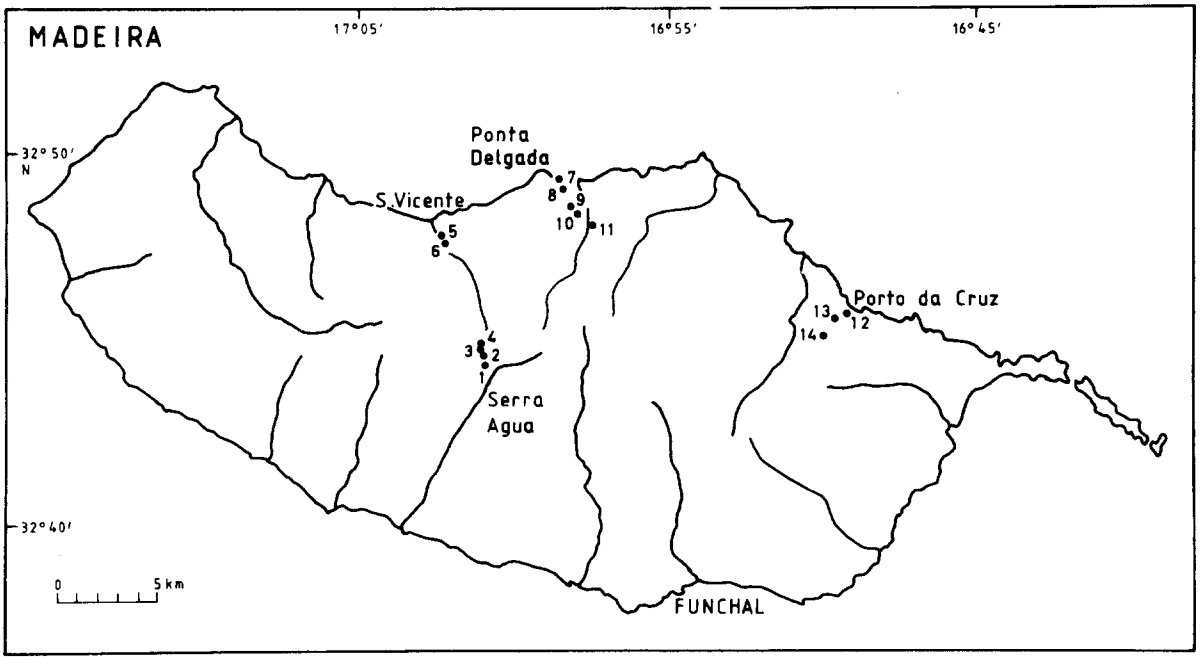


Fig. 2. Distribution of palaeomagnetic sampling sites.

i.e., lavas (visibly altered) in association with thick tuff layers, and minor intrusives. Sites 1–4 are cut by numerous near-vertical dykes. In the other sampling regions (Ponta Delgada and Porto da Cruz, see Fig. 2) the rocks are mineralogically much fresher, and for this reason some of the sites, notably nos. 9 and 13, are strongly suspected to represent Neogene intrusions. With these exceptions the sampled material is likely to constitute rocks that are older than those investigated by Watkins et al. (1966) and Watkins (1973). From the total rock collection altogether 140 specimens have been studied.

Sun compass orientations are available for 67% of the collected samples. The remaining material has been oriented magnetically, sometimes combined with sightings of pronounced topographic features. Table I gives further sampling details.

3. Experimental results

The specimens have been measured on a shielded Digico Spinner Magnetometer and treated by progressive demagnetization by means of altering field and/or temperature. Eighty nine of these specimens define stable end point magnetization. To estimate remanence components other than high stability ones analysis by vector subtraction or orthogonal vector diagrams (Zijderveld, 1967) have been attempted on all specimens defining great circle trends on demagnetization. However, from this latter category of samples only four allowed estimation of an additional component of magnetization. Some specimens are magnetically unstable throughout the laboratory treatment and are consequently set aside in the further discussion. Others decay into the noise level before a terminal direction is reached. Special attention has been made to identify specimens that pick up significant viscous magnetization at some stages of laboratory treatment.

TABLE I

Details of palaeomagnetic collection (see Fig. 2 for site locations)

Site no.	Sample nos.	Rock type, Formation etc.
1	MA 1– 3	Old volcanic complex. Lava transected by numerous younger and near-vertical dykes
2	4– 6	——————,,——————
3	7– 9	——————,,——————
4	10–12	——————,,——————
5	13–17	Granitic/syenitic intrusion of the 'basement' complex
6	18–20	——————,,——————
7	21–23	Lava in 'basement' complex
8	24–26	
9	27–29	Due to the unusually fresh appearance of this rock it is probably a Neogene intrusion
10	30–32	
11	33–35	
12	36–39	
13	40–42	Probably a Neogene intrusion (for the same reason as for site 9)
14	43–45	

Figures 3–5 give a representative coverage of demagnetization results, particularly for specimens that are included in the final palaeomagnetic data. Altogether the rock collection defines three axes of magnetization. The most predominant remanence component, the A-axis, has a roughly N–S declination and intermediately steep inclination. This dual polarity axis which predominates in higher site numbers is exclusively defined by stable end point data; for individual specimen results see for example MA8-A1 (Fig. 3), MA10-A1 (Fig. 4), MA28-A1 (Fig. 3) and MA30-B1 (Fig. 4). The remaining two axes are both practically flatlying, the B-remanence being mostly directed SSW, while the C-magnetization has WNW and ESE declinations. The multicomponent nature of the magnetic remanence is particularly pronounced in the lower site numbers (nos. 1–6) where various component combinations occur (A normal plus A reversed plus B normal in site 1, A normal plus C normal in site 2, etc.).

The 92 individual directions (from 89 specimens) that are regarded to provide palaeomagnetic information are listed in Table II. The fairly stable bulk magnetization of sites 5 and 6 are anomalous (with respect to the other results of this study and to what might be anticipated for relative Mesozoic/Cainozoic field directions) and have not been accepted as relevant palaeomagnetic markers. The remanence intensity, J_n, of these two sites are a

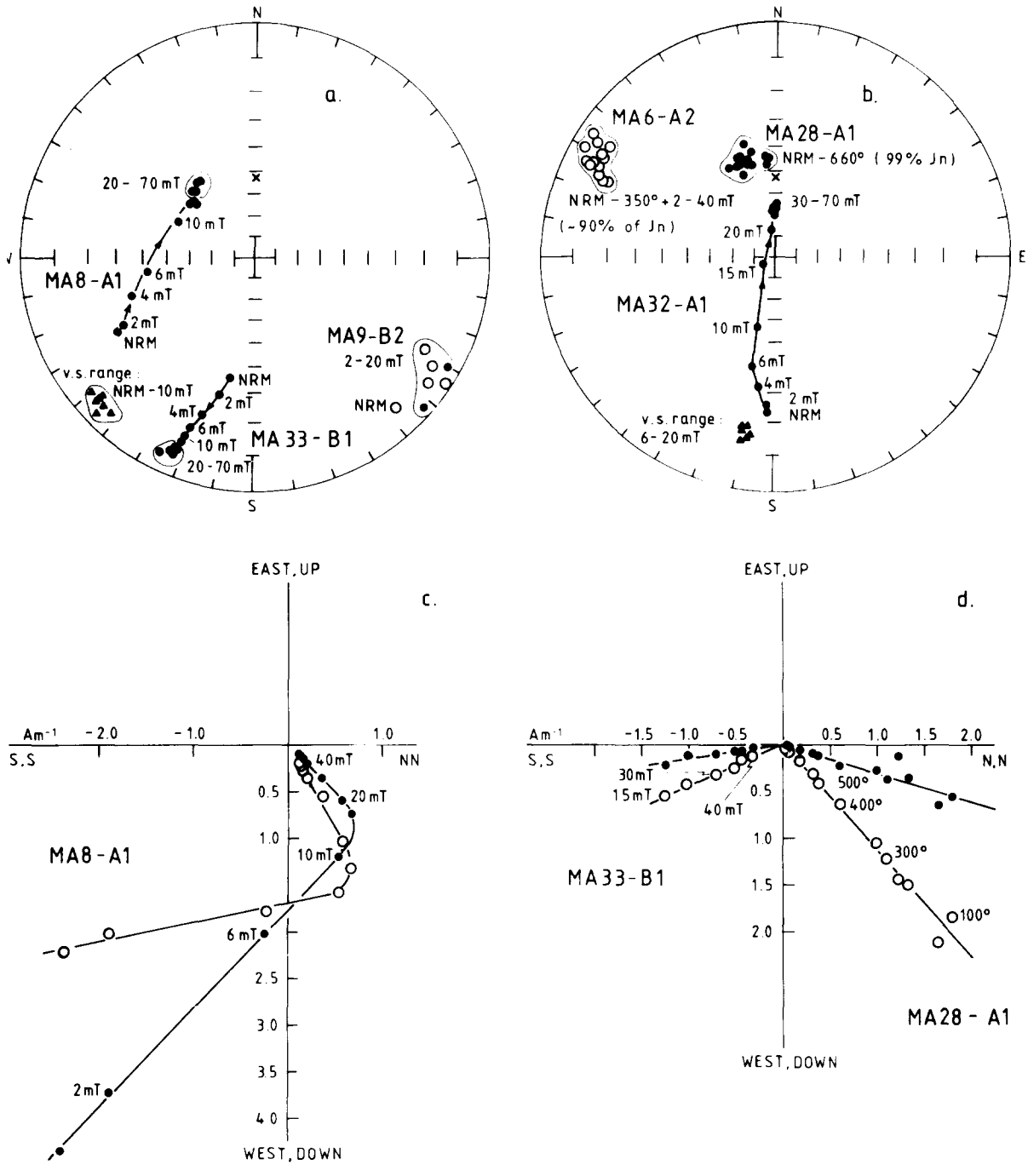


Fig. 3. Examples of demagnetization results. The data are either given as directional behaviour plotted in stereonets (a and b) or as orthogonal vector projections (c and d). In the stereoplots the closed (open) symbols represent downward (upward) pointing remanence directions. For the vector diagrams open (closed) symbols are points in the vertical (horizontal) plane (note that the horizontal plane is inverted). The intensity unit is $1 \cdot Am^{-1}$. Thermal demagnetization steps are marked by the temperature in °C while alternating field data are indicated by the peak field value in mT. Vector subtracted directions (MA8-A1 and MA 32-A1) are shown by triangles. (See text for further details.)

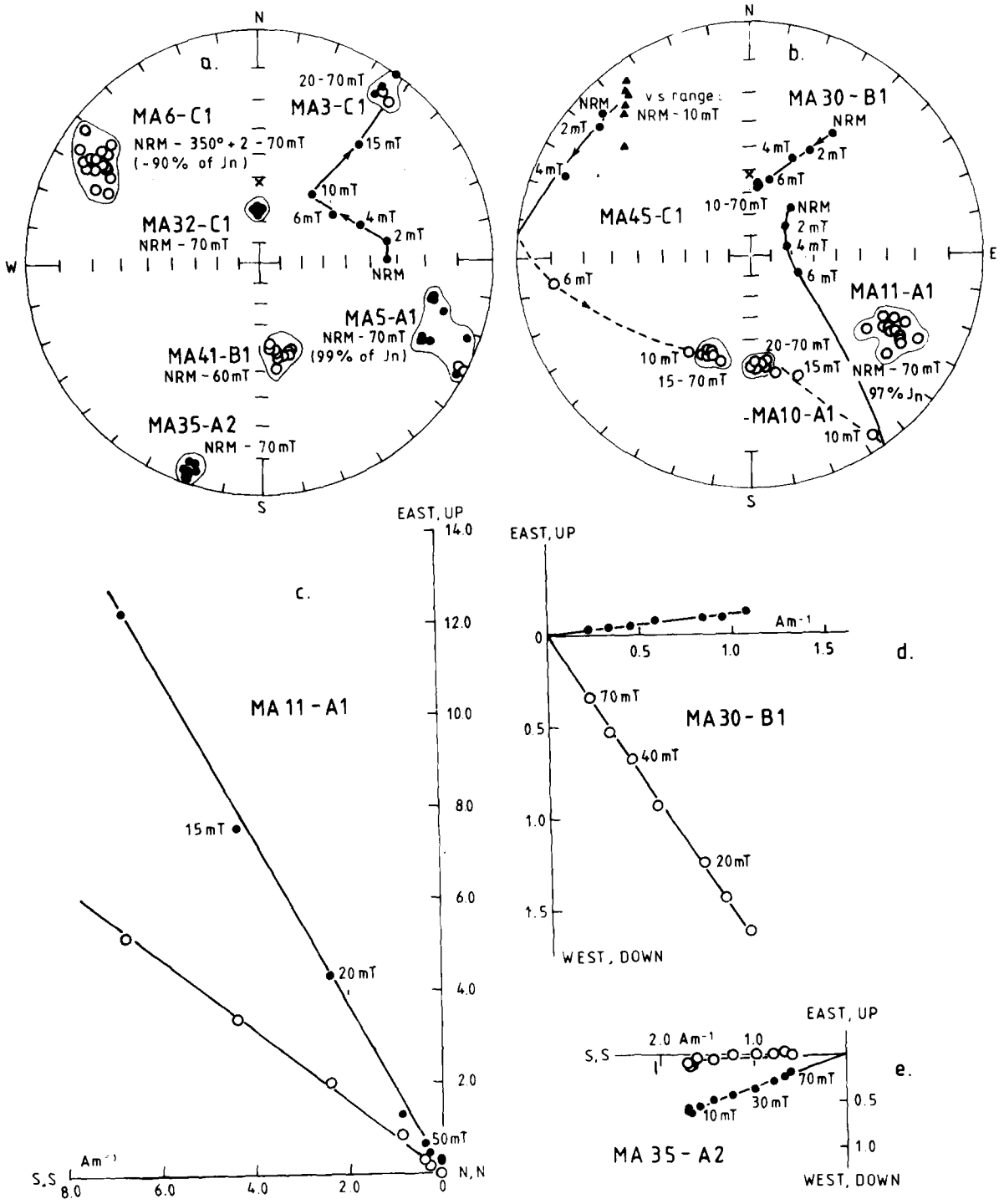


Fig. 4. Further demagnetization data. Diagram conventions as for Fig. 3. J_n denotes the intensity of natural remanent magnetization (NRM). (Further explanations in the text.)

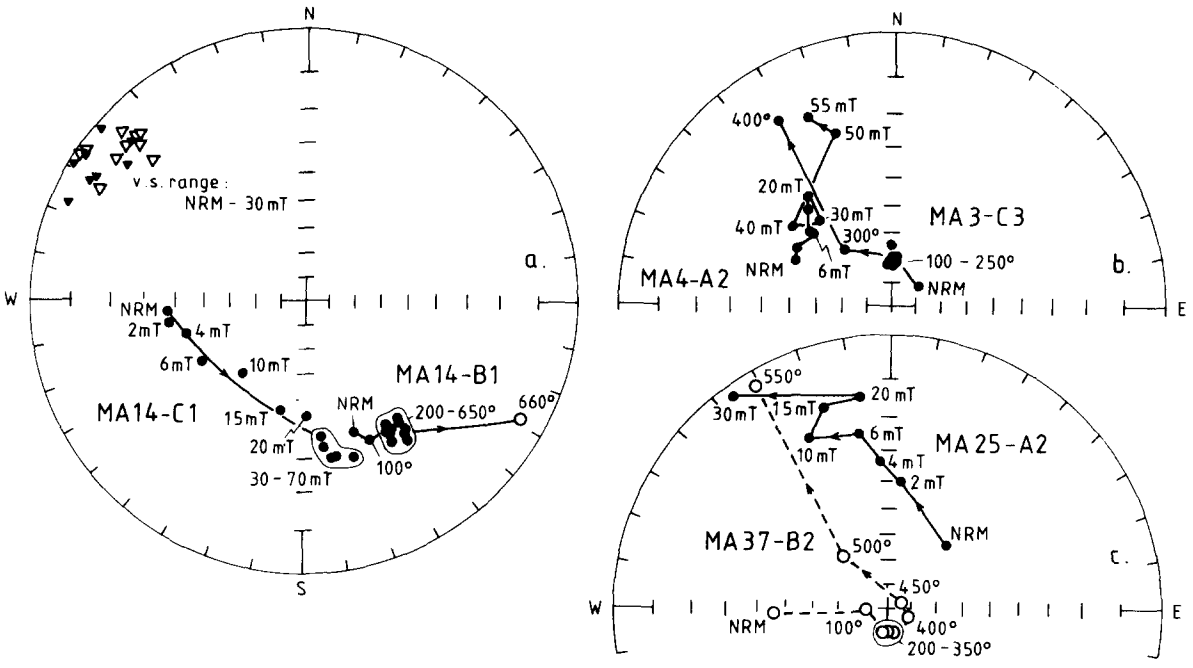


Fig. 5. Further demagnetization results.

factor 10^3 below that for the remaining sites ($\sim 5 \cdot 10^{-3}$ Am^{-1} as compared to $\sim 5 Am^{-1}$). Most specimens from sites 5 and 6 were hardly affected by AF treatment up to 70 mT. Thermal demagnetization shows that there is a very narrow blocking temperature (T_B) range just above 650°C and which accounts for the bulk of the intensity of magnetization. Associated with the rapid J_n-drop the magnetization directions tend to move into the C-axis positions at 650°C, but unfortunately the intensity decays into the noise level before the C-axis directions can be experimentally verified. Specimens MA14-B1 and MA14-C1 (Fig. 5) give directional data from site 5. AF demagnetization of MA14-C1 below 30 mT erases the normal C-axis magnetization but the directional stabilization above 30 mT is in an anomalous position. However, the high blocking temperature range suggests haematite as the important magnetic carrier. It therefore appears reasonable to suppose that the natural remanent magnetization (n.r.m.) of this site is of chemical origin, being acquired along the C-axis in a time span long enough to record polarity inversion(s) (i.e., the anomalous bulk magnetization is the resultant of normal and reverse C-axis components). The same explanation is invoked for site 6 which has steep upward pointing bulk directions with a certain smear in the plane of the C-axis.

Figure 5b and c show directional data for some other specimens that are not included in Table II (i.e., the high field/high temperature component can not be satisfactorily established). Nevertheless these results give further support to the existence of the C-magnetization.

A few specimens, from sites 3, 5 and 11, have been subjected to further laboratory analysis by application of reflection microscopy, saturation magnetization versus temperature, J_s–T, and measurements of the remanence coercive force, H_{CR}. Site 5 material is too weakly magnetic to provide meaningful J_s–T data, and the opaque grains are too small for microscopic identification. However, the relatively high H_{CR}-values (cf. MA14, Fig. 6c)

TABLE II

Palaeomagnetic results from Madeira. Most directions represent stable end point results. By vector subtraction (v.s.) four samples have given palaeomagnetic components other than the high field/high temperature ones

Site	Specimen	D	I	Group	Range
1	MA 1 B1	002	50	A	6–70 mT
	1 C2	354	64	A	NRM-300° + 2–30 mT
	2 B1	181	−32	A	6–70 mT
	2 B3	152	−41	A	100–300° + 4–40 mT
	3 C1	036	01	B	20–70 mT
2	5 A1	112	11	C	NRM-70 mT
	5 A2	338	35	A	300° + 2–15 mT
	5 A3	345	35	A	NRM-350°
	6 A2	299	−09	C	NRM-350° + 2–40 mT
	6 B1	298	−12	C	NRM-70 mT
	6 B2	298	−12	C	NRM-70 mT
	6 C1	302	−12	C	NRM-350° + 2–70 mT
3	8 A1a	227	07	B	v.s. NRM-10 mT
	b	319	48	A	20–70 mT
	8 B1	314	44	A	2–50 mT
	8 B2	325	43	A	NRM-350° + 2–70 mT
	9 A1	122	−08	C	4–15 mT
	9 A2	119	−05	C	4–50 mT
	9 B2	124	−03	C	2–20 mT
	9 C1	110	−23	C	20°–500°
4	10 A1	178	−40	A	30–70 mT
	10 A2	174	−42	A	NRM-40 mT
	10 B1	186	−46	A	40–60 mT
	11 A1	120	−20	C	NRM-70 mT
	11 A2	125	−18	C	NRM-350° + 2–55 mT
5	14 C1	305	−02	C	v.s. NRM-30 mT
	14 C2	350	33	A	NRM-350° + 2–70 mT
7	23 A1	354	27	A	10–50 mT
	23 A3	001	26	A	10–60 mT
	23 B1	346	34	A	10–50 mT
	23 B3	341	24	A	10–50 mT
8	24 A1	201	−55	A	15–70 mT
	24 B1	175	−51	A	100–400°
	24 B2	205	−66	A	20–70 mT
9	27 A1	357	43	A	6–60 mT
	27 A2	355	40	A	NRM-70 mT
	27 B1	357	39	A	15–40 mT
	27 B2	001	43	A	2–70 mT
	27 B3	357	41	A	15–40 mT
	27 C1	357	37	A	100–450°
	27 C2	355	36	A	10–40 mT
	28 A1	344	45	A	NRM-660°
	28 A2	329	42	A	NRM-40 mT
	28 B1	340	53	A	10–70 mT
	28 B2	339	51	A	NRM-70 mT
	28 B3	338	53	A	NRM-70 mT
	28 B4	336	55	A	NRM-70 mT
	29 B1	344	44	A	4–70 mT
	29 Cl	339	44	A	4–70 mT
10	30 A1	006	55	A	NRM-70 mT
	30 B1	006	56	A	10–70 mT
	30 B2	011	56	A	NRM-660°

TABLE II (continued)

Site	Specimen	D	I	Group	Range
	30 C1	006	56	A	NRM-70 mT
	31 A1	011	55	A	NRM-70 mT
	31 A2	012	56	A	NRM-550°
	31 A3	024	60	A	NRM-70 mT
	31 B1	005	59	A	40–70 mT
	31 B2	025	61	A	NRM-70 mT
	31 B3	009	58	A	NRM-70 mT
	32 A1a	190	16	B	v.s. 6–20 mT
	b	357	67	A	30–70 mT
	32 A2	010	67	A	NRM-70 mT
	32 B2	001	62	A	NRM-70 mT
	32 C1	359	64	A	NRM-70 mT
	32 C2	015	69	A	NRM-70 mT
11	33 A1	199	08	B	NRM-580°
	33 A2	201	09	B	NRM-70 mT
	33 B1	203	07	B	15–70 mT
	33 B2	205	06	B	NRM-70 mT
	33 B3	203	09	B	NRM-70 mT
	35 A1	200	01	B	NRM-600°
	35 A2	200	02	B	NRM-70 mT
	25 A3	200	03	B	NRM-70 mT
	35 B1	200	03	B	NRM-70 mT
	35 B2	200	05	B	NRM-70 mT
13	40 A2	176	−45	A	NRM-40 mT
	40 B1	188	−51	A	NRM-40 mT
	40 C1	191	−42	A	NRM-30 mT
	41 A1	179	−53	A	400–550°
	41 A2	159	−45	A	NRM-60 mT
	41 B1	169	−46	A	NRM-60 mT
	41 B2	171	−39	A	NRM-40 mT
	42 A1	169	−37	A	NRM-40 mT
	42 B1	171	−42	A	NRM-50 mT
14	44 B1	307	26	C	500–580°
	44 B2	168	−58	A	NRM-40 mT
	44 C1	186	−61	A	15–70 mT
	45 A2	194	−53	A	2–70 mT
	45 B1	201	−46	A	15–70 mT
	45 C1a	321	10	C	v.s. NRM-6 mT
	b	202	−41	A	15–70 mT
	45 C2	194	−56	A	2–70 mT

support the conclusion based on thermal demagnetization (Fig. 6b) that the important magnetic mineral in this site is haematite.

Site 11 shows an excessively altered mineralogy of low temperature type; the grains are practically 100% granulated. The original magnetic minerals were titanomagnetites but the present magnetic phases are 'magnetite' (owing to the extensive low temperature alteration the actual mineral phase may well be a cation-deficient one) and haematite. The presence of 'magnetite' is demonstrated by Curie temperatures around 550°C and by low H_{CR} values. The groundmass has a reddish colour.

Of the three sites investigated no. 3 has apparently the least altered oxide mineralogy. The opaque grains are relatively fresh titanomagnetite and ilmenite particles, but the tendency of the titanomagnetites to be white coloured, particularly

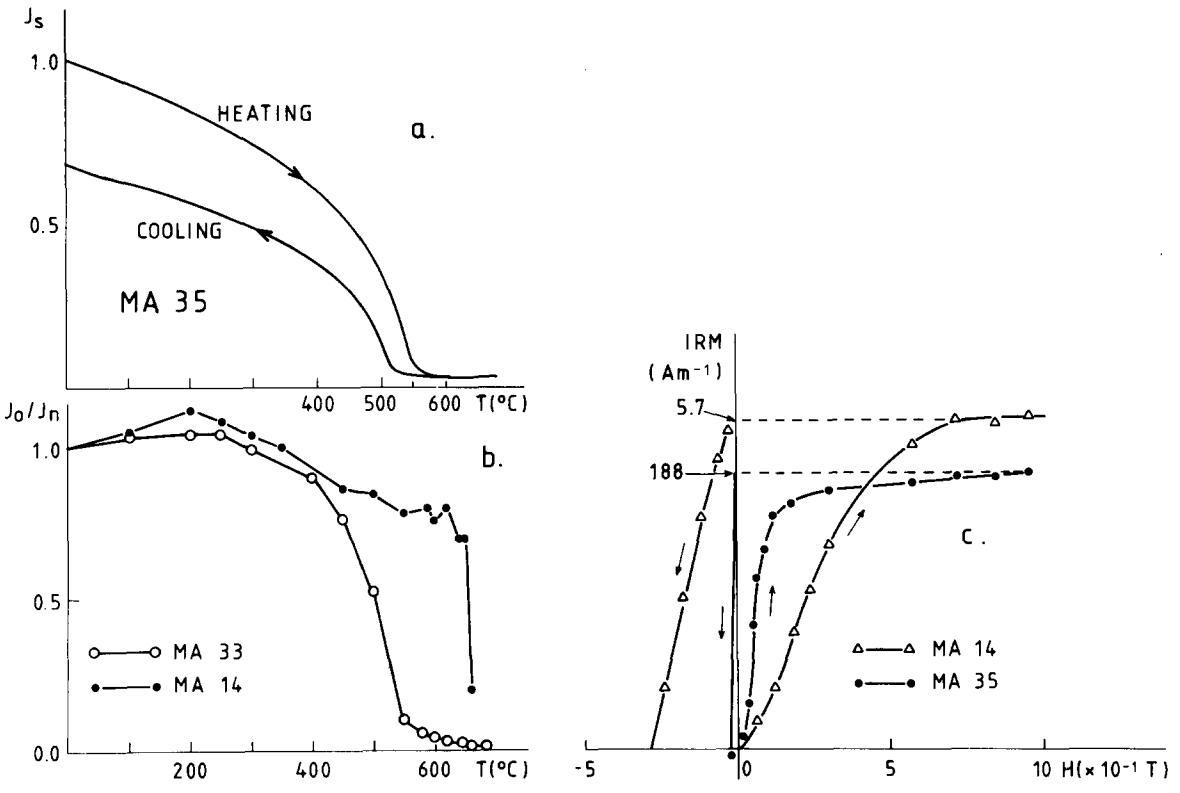


Fig. 6. Some rock magnetic properties for sites 5 (MA 14) and 11 (MA 33 and MA 35). Diagram c shows examples of remanence coercive force (H_{CR}) values from the two sites. The low H_{CR} for site 11 tends to be governed by a magnetite–maghaemite mineralogy, as suggested by measurements of saturation magnetization (J_S) versus temperature (diagram a) and the thermal decay pattern of the natural magnetic moment (diagram b). Note that the high H_{CR} for site 5 is associated with a haematite blocking temperature range (b).

along cracks and rims and when associated with a red groundmass, suggests that site 3 has also been affected by low temperature alteration. These observations are therefore fully compatible with the multicomponent nature of the remanent magnetization as observed in most sites of the present study.

4. Interpretation

In order to illustrate the component parts of the remanence (axes A–C) the magnetization directions of Table II are plotted in Fig. 7. Table III provides the overall palaeomagnetic parameters.

In a case like this, where the rocks frequently display multicomponent magnetization, it would on the whole be inadequate to consider specimen directions as palaeomagnetic "spot"-readings. The suggested acquisition of chemical magnetization (following from the relatively high degree of low temperature alteration of the original iron–titanium oxides) within a rock unit (lava or intrusive) is certainly a function of micro-environmental oxidizing conditions which in turn are time-dependent, i.e., various parts of even a small rock fragment may have become oxidized (and remagnetized) in distinctly different periods of time. Hence, individual specimen directions are likely to constitute a certain averaging-out of ancient geomagnetic secular variation. For the statistical treatment it is more realistic therefore to give unit

weight to individual specimen directions rather than to site mean directions. Exceptions may be sites 9 and 13, which probably represent younger intrusives (in which the original TRM may still be predominant), but in the present case a varied weighing procedure would have no significance for the conclusions.

Axis A matches closely with the direction of the present geocentric dipole field and accords well with the Late Tertiary palaeomagnetic field directions from Madeira, as established by Watkins et al. (1966) and Watkins (1973). This directional concordance is not surprising in that at least two of the A-axis sites are likely to represent Late Tertiary intrusives, and in addition one might indeed anticipate at least partial remagnetization of

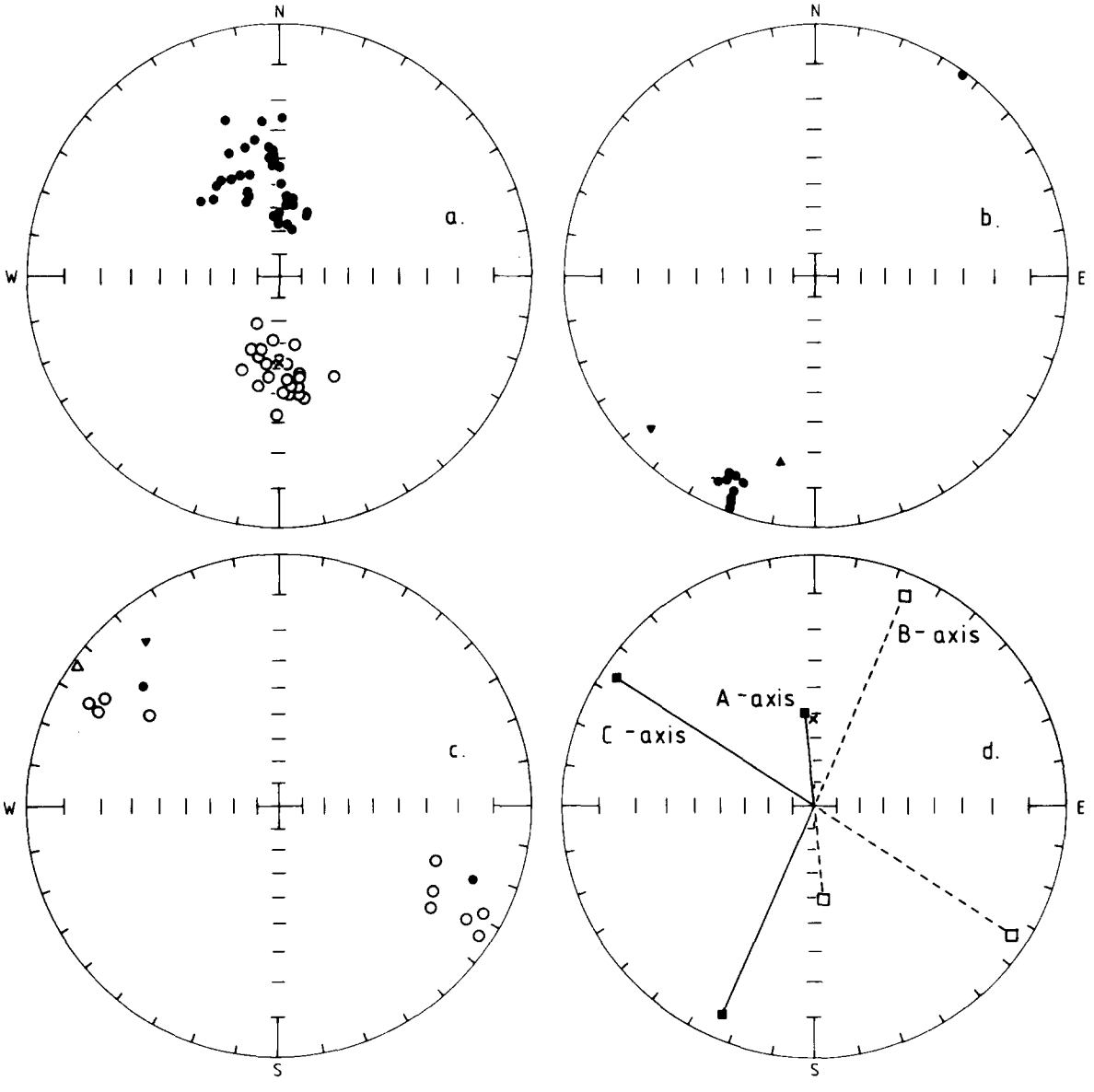


Fig. 7. Palaeomagnetic groupings, based on individual specimen results (Table II), showing axes A–C.

TABLE III

Mean palaeomagnetic data from Madeira

Group	N	R	K	α_{95}	$\bar{D}$	$\bar{I}$	Pole	δ_p	δ_m
A	64	61.9	30.3	3.2	355.1	49.3	042.0E, 85.1S	2.8	4.2
B	13	12.8	66.4	4.8	203.4	5.8	306.7E, 48.0S	2.4	4.8
C	14	13.5	26.6	7.3	301.6	3.9	056.3E, 27.3S	3.7	7.3
I *	30	29.2	36.3	4.2	002.0	47.4	320.3E, 85.4S	3.5	5.5

N = number of unit vectors (specimens); R = length of resultant; K = precision parameter; α_{95} = radius of circle of confidence at 95% significance level; $\bar{D}$, $\bar{I}$ = declination, inclination of mean vector; δ_p, δ_m = semi-axes of VGP oval of confidence at 95% significance level. Statistics according to Fisher (1953).

* Combined Pleistocene and Pliocene data on extrusives from Madeira (Watkins, 1973).

any pre-Neogene rock owing to the thermo–chemical processes that are likely to have accompanied the extensive Neogene volcanism on the island. Contrary to the A-magnetization, the B and C components are significantly different from the Late Tertiary palaeomagnetic reference field. There appears to be no general difference in remanence stability for the three components of magnetization, and in all three groups one can find specimens that are stable over the entire range of demagnetization. Furthermore, both the B and the C group represent samples that have very different field orientations. This random orientation procedure along with the consistent remanence groupings (after field corrections) probably provides the strongest arguments for regarding the B and C magnetizations as true palaeomagnetic field markers. Also, based on the available geological information there is no basis for suggesting that the B and C magnetizations are the results of tectonic rotation of an original A magnetization.

Madeira lies south of the Azores–Gibraltar fracture zone, which is considered as the Africa/Europe plate boundary, so any palaeomagnetic comparison with pre-Neogene times should be with reference to Africa. As seen from Fig. 8 the palaeomagnetic pole relative to Africa remained fairly stationary from Triassic to ca. mid-Cretaceous (Albian/Senomanian), but at around Barremian/Aptian times (upper Lower Cretaceous) an excursion to a ca. 20° lower palaeolatitude appears to have occurred (Bardon et al., 1973; Gidskehaug et al., 1975). From ca. Albian/Senomanian to Lower Tertiary a westward polar migration, roughly along the 60°S parallel, is envisaged.

Neither the B nor the C pole is in full agreement with the post-Palaeozoic polar curve of Africa. However, our B and C magnetizations may not be sufficiently precisely defined for a perfect comparison. As seen from Fig. 7c the two polarity groups of the C-remanence are not entirely antiparallel (ca. 160° apart), a feature that may be the result of an unresolved interplay between the normal and reverse C-axis components. In fact, an overall axis change of only some 15° in inclination (in a downward sense for the normal component) is sufficient to bring the C-pole into proper agreement with the upper Lower Cretaceous poles for Africa. Also, the rock sequence is described as being practically undisturbed tectonically, but only a modest tilting is required to bring about the observed polar divergence.

The Madeira B-pole falls roughly in the extension of the Upper Cretaceous polar pattern, ca. 50° of longitude to the west of the suggested knee in the relative polar wander curve (Fig. 8). The B magnetization is essentially based on results from only one site (no. 11) so factors like local tectonism and inadequate time representation in the palaeomagnetic record (for cancelling out geomagnetic secular variation) may possibly explain the observed branch off in pole location. However, the suggestion of the pole B being a virtual palaeomagnetic pole is hardly supported by the mineralogical evidence; the extensive low temperature alteration of the iron–titanium oxides of this site may suggest lengthy magnetization processes with associated averaging effects on palaeomagnetic secular variation. Nevertheless, pole B should be regarded as much less reliable than pole C. Despite these palaeomagnetic matching problems

there appears hardly any other alternative at present than to assigne Cretaceous–? Lower Tertiary ages to the C and B magnetizations.

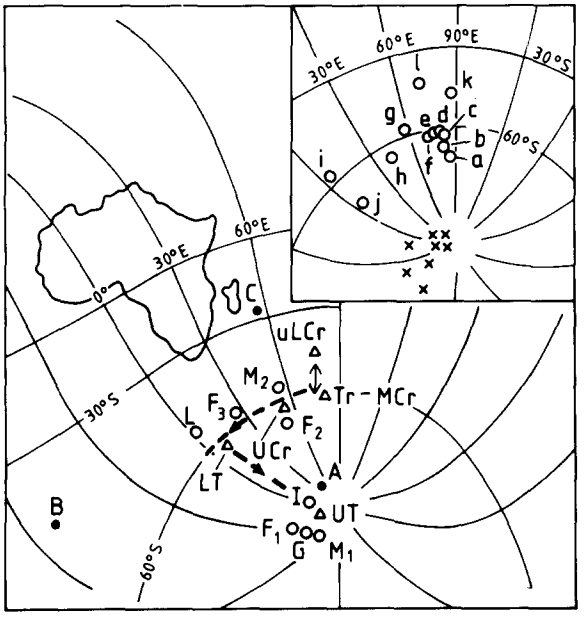


Fig. 8. Pole positions based on the Madeira A, B and C magnetizations (solid symbols) in conjunction with assumed relevant reference data for Africa and from east central Atlantic islands. Poles G, F_1, I and M_1 represent the Neogene of Gran Canaria/Tenerife (Storetvedt et al., 1978), Fuerteventura (Storetvedt et al., 1979), Madeira (Watkins et al., 1966; Watkins, 1973) and Maio (Storetvedt and Løvlie, 1983), respectively. Poles F_2 and F_3 are from the early subaerial volcanics of Fuerteventura (Storetvedt et al., 1979), M_2 is from the Cretaceous volcanics of Maio (Storetvedt and Løvlie, 1983) and pole L is from the lava Series I of North Lanzarote (Johansen, 1976). In the inserted diagram crosses are African Tertiary (mostly Upper Tertiary) data and the lettered poles refer to the following results: a) Mlanje (Briden, 1967), b) mean Mesozoic SE Africa (Hailwood and Mitchell, 1971), c) Lupata, 106 Ma (Gough and Opdyke, 1963; Gough et al., 1964), d) Shawa ijolite (Gough and Brock, 1964), e) mean Mesozoic NE. Africa (Hailwood and Mitchell, 1971), f) Hoachanas (Gidskehaug et al., 1975), g) Kimberlites, 83–89 Ma (McFadden and Jones, 1977), h) Wadi Natash volcanics, 81–90 Ma (El Shazly and Krs, 1973), i and j) Lower–(Upper) Tertiary volcanics Egypt (Gouda Hussain et al., 1979), k) Kaoko lavas, 110–128 Ma (Gidskehaug et al., 1975) and l) Lower Cretaceous, Atlas Mountains (Bardon et al., 1973). Triangle symbols are mean poles for the following groups: the crosses, UT (Upper Tertiary), a–f, Tr–MCr (Triassic–mid-Cretaceous), g and h, UCr (Upper Cretaceous), i and j, LT (Lower Tertiary), and k and l, u LCR (upper Lower Cretaceous), respectively.

The palaeomagnetic evidence for Cretaceous volcanic activity in Madeira gains corroborative support from DSDP site 136 (Hayes et al., 1972) located in an area of abyssal hills only 160 km north of the island (Fig. 1). The drill bottomed in tholeiitic basalt underlying late Aptian marls but early Aptian shales were recovered from bit cuttings. If the basalt encountered is of extrusive origin it has a minimum age of ca. 110 Ma. However, definite evidence for Cretaceous volcanism is provided by the three ash horizons of this site, being interstratified with sediments ranging in age from late Aptian–Coniacian/Santonian (i.e., ca. 105 Ma–ca. 85 Ma). This shows indeed that the Madeira region was volcanically active in Lower/Middle Cretaceous times.

We conclude therefore that the inferred oldest palaeomagnetic component of this study (the C-magnetization) dates back to the late Lower Cretaceous and may correspond to the earliest volcanic evolution of the island. These 'basement' volcanics have subsequently been affected by at least two magnetization pulses, the first one (of minor importance) in the Late Cretaceous–? Early Tertiary (the B-component) and the second and dominating one, the A-remanence, during the extensive Late Tertiary volcanism in the area. Cretaceous magnetization with a strong Neogene overprint has also recently been observed in the Cretaceous volcanics of Maio, Cape Verde Islands (Storetvedt and Løvlie, 1983), and similar results have previously been reported from the Canary Islands (e.g., Storetvedt et al., 1979). Though future studies should aim at establishing a broader data base for the B- and C-magnetization the total information already available makes it unlikely that the C-component will change significantly. Indeed, there is now quite strong evidence for there having been basically two stages of magnetic activity in the volcanic islands off NW Africa, i.e., in Late Mesozoic and Late Tertiary times, respectively.

Acknowledgements

The financial support to M.C. Abranches from Invotan and Fundacao C. Gulbenkian is gratefully acknowledged. The field expenses were covered by a grant from Kr. Birkelands Fond.

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